

STATS 507 – Data Science Analytics using Python

Project Proposal Guideline (10 pts)

In the project proposal, you'll pick a project idea to work on early.

General Guideline

- Your project must intergrade with at least one Hugging Face **model** or one open **dataset**.
- Each project should be done individually and must be implemented in Python.
- You should scope your project to be about what you could complete within the time constraints.
- The final project should be submitted as a GitHub repository containing a well-documented executable codebase along with a 2-page summary report (summary.md or pdf) that concisely describes your problem, approach, and results. You can have preliminary results in your proposal but choose a problem that is non-trivial and not a Ph.D. thesis either.
- You are expected to spend at least 40 hours on your project.

Potential Paths

- Dataset-Centric Projects: choose an open dataset that we didn't cover in class for meaningful analysis. <https://huggingface.co/docs/datasets/en/index>
- Model-Centric Projects: choose a model/models from Hugging face hub to build innovative applications. <https://huggingface.co/models>
- Many more!

Proposal Requirements

- Format: at least 2 pages, at least 1 figure.
- Sections:

Overview

- Describe the background and motivation of the project.
- Why would you want to do this project?
- What kind of data/model do you want to work on?
- What kind of insights are you expecting from this projects?

Prior Work

- Literature review: conduct a comprehensive review of existing literature, focusing on recent advancements and state-of-the-art.
- Describe potential methods that can be used to achieve the project goal.

Preliminary Results

- Data understanding (dataset statistics, data quality assessment, initial insights, potential challenge)
- Basic model (resource requirements, performance bottlenecks...)
- What tools from class did you use, what tools will you explore?

Project Deliverables

- What will a successful project produce?
- What are your sub-goals?

Timeline

- Week 1-2
- Week 3-4
- Week 5